

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes Division

Cluster Programme - M. Tech in AI & ML

I Semester , 2023 – 24(January,2024)

Mid semester Examination (**MAKEUP**)

Course No : AIMLC ZC418
Course Title : Introduction to Statistical Methods
Nature of Exam. : Closed Book (Online)
Weightage : 30 Marks
Duration : 120 minutes
Date : 03rd February,2024_FN

Number of questions:4

Number of Pages: 4

Q. No	Question	Marks
Q.1.a)	Discuss at least three statistical concepts which helps to understand the data as a part of pre-processing. Data Cleaning: Data cleaning is a crucial step in the data preprocessing pipeline, involving the identification and correction of errors or inconsistencies in a dataset. This process ensures that the data is accurate, complete, and ready for analysis. Common tasks include handling missing values, removing duplicates, and addressing outliers that may distort the overall picture. Data Integration: Data integration is the process of combining data from different sources to provide a unified view. It involves merging datasets with varying structures, formats, and semantics into a cohesive and comprehensive dataset. Successful data integration enhances the overall quality of the data and enables more meaningful analysis and insights. Data Transformation: Data transformation involves converting and restructuring data to meet the requirements of the analytical process or to comply with a specific data model. This step may include normalization, encoding categorical variables, or scaling numerical features. Data transformation aims to enhance the compatibility and usability of the data for subsequent analysis. Data Reduction: Data reduction involves simplifying and summarizing large datasets while retaining essential information. Techniques like dimensionality reduction, aggregation, or sampling are employed to reduce the dataset's size without significantly sacrificing its representativeness. This step is essential for improving computational efficiency and reducing the risk of overfitting in machine learning models. Importing Crucial Libraries: To streamline the data preprocessing tasks, various libraries in programming languages like Python are essential. Commonly used libraries include Pandas for data manipulation, NumPy for numerical operations, and Scikit-learn for machine learning functionalities. These libraries provide powerful tools to efficiently handle and analyze data. Identifying and Handling Missing Values: Missing values are a common challenge in datasets and can adversely impact analysis and model performance. Identifying and handling missing values involves techniques such as imputation, deletion, or using advanced algorithms to estimate missing values. The goal is to maintain data integrity and avoid bias in subsequent analyses.	3 M

Q.1.b)	Consider two events such that $P(A) = 0.6$ and $P(B) = 0.7, P(A \text{ and } B) = 0.3$. Then find the following probabilities of the events mentioned i) Only A ii) Only B iii) A and B^c iv) A^c and B (Where superscript c indicates the complimentary event) <u>Sol:</u> $P(A)=0.6, P(B)=0.7, P(A \text{ and } B)=0.3$ - i). only A = $P(A) - P(A \text{ and } B) = 0.6 - 0.3 = 0.3$ ans 1mark - ii). Only B = $P(B) - P(A \text{ and } B) = 0.7 - 0.3 = 0.4$ ans 1mark - iii). A and $B^c = P(A) - P(A \text{ and } B) = 0.6 - 0.3 = 0.3$ ans 1mark - iv). A^c and B = $P(B) - P(A \text{ and } B) = 0.7 - 0.3 = 0.4$ ans 1mark	4 M
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Q.2.a

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Let A and B be two mutually exclusive events such that $P(A) = 0.45$, $P(B) = 0.20$. Then find

3 M

i) $P(B|A \cup B)$ ii) $P(A|A \cap B)$ iii) $P(A^c|A \cup B)$

iv) $P(A^c|A \cap B)$

2Q. (a) Given $P(A) = 0.45$, $P(B) = 0.2$

and A & B are M.E. events

i.e. $P(A \cup B) = P(A) + P(B)$

$$= 0.45 + 0.2 = 0.65$$

and $P(A \cap B) = 0$

$$(i) \quad P(B|A \cup B) = \frac{P[B \cap (A \cup B)]}{P(A \cup B)}$$

$$= \frac{P(B)}{P(A \cup B)} = \frac{0.2}{0.65} = 0.3077$$

$$(ii) \quad P(A|A \cap B) = \frac{P[A \cap (A \cap B)]}{P(A \cap B)}$$

$\therefore$ The Probability doesn't exist [$\because P(A \cap B) = 0$]

$$(iii) \quad P(A^c|A \cup B) = \frac{P[A^c \cap (A \cup B)]}{P(A \cup B)}$$

$$= \frac{P(B)}{P(A \cup B)} = \frac{0.2}{0.65} = 0.3076$$

$$(iv) \quad P(A^c|A \cap B) = \frac{P[A^c \cap (A \cap B)]}{P(A \cap B)}$$

Probability doesn't exist [$\because P(A \cap B) = 0$]

Q.2.b)	A product is manufactured by four companies A, B, C and D. They produce 20%, 30%, 40% and 10% of the products released in the market. But it is observed that 30%, 20%, 10% and 25% are defective items manufactured by them respectively. Then find - Total percentage of defective items manufactured - The percentage that the selected defective item is produced by company A - The percentage that the selected defective item is not produced by company B - The percentage that the selected defective item is not produced by company D <u>(2).b</u> Given $P(A)=0.2$, $P(B)=0.3$, $P(C)=0.4$, $P(D)=0.1$ Let x = defective item $P(x/A)=0.3$; $P(x/B)=0.2$; $P(x/C)=0.1$ and $P(x/D)=0.25$ (i) $P(x) = P(A)P(x/A) + P(B)P(x/B) + P(C)P(x/C) + P(D)P(x/D)$ $= (0.2)(0.3) + (0.3)(0.2) + (0.4)(0.1) + (0.1)(0.25)$ $= 0.185 = 18.5\%$ (ii) $P(A/x) = \frac{P(A)P(x/A)}{P(x)}$ $= \frac{0.06}{0.185} = 0.3243$ $= 32.43\%$ (iii) $P(B^c/x) = 1 - P(B/x)$ $= 1 - \left[\frac{P(B)P(x/B)}{P(x)} \right]$ $= 1 - \left[\frac{0.06}{0.185} \right] = 0.6757 = 67.57\%$ (iv) $P(D^c/x) = 1 - P(D/x)$ $= 1 - \left[\frac{0.025}{0.185} \right]$ $= 0.8649 = 86.49\%$	4 M
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Q.3.a)	Consider a random variable X that follows Poisson distribution with mean as 4 then find the probability that i) X takes a value less than 5 ii) X takes a value greater than 5 i) X lies between 2 and 5 Probability mass function of Poisson distribution is $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$ i) X takes a value less than 5 $P(X < 5) = P(0) + P(1) + P(2) + P(3) + P(4)$ $= 0.629$ ii) X takes a value greater than 5 $P(X > 5) = 1 - P(X \leq 5)$ $P(X > 5) = 1 - [P(0) + P(1) + P(2) + P(3) + P(4) + P(5)]$ $= 1 - 0.785$ $= 0.215$ iii) X lies between 2 and 5 $P(2 < X < 5) = P(3) + P(4)$ $= 0.1954 + 0.1954$ $= 0.3908$	3 M

Q.3.b

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Consider the following probability distribution.

$$f(x, y) = \frac{k}{2} (e^{-(x+y)}), 0 < x < \infty, 0 < y < \infty$$

- i) Find k value
- ii) Check whether x and y are independent or not.
- iii) Find $P(0 < x < 2, 0 < y < 5)$
- iv) $P(0 < x < 2 / 0 < y < 5)$

5 M

$$\int_0^{\infty} \int_0^{\infty} \frac{k}{2} e^{-(x+y)} dx dy = 1$$

$$\int_0^{\infty} \left(\frac{k}{2} e^{-y} \int_0^{\infty} e^{-x} dx \right) dy = 1$$

$$\int_0^{\infty} \left(\frac{k}{2} e^{-y} (0 - (-1)) \right) dy = 1$$

$$\int_0^{\infty} \frac{k}{2} e^{-y} dy = 1$$

$$\frac{k}{2} [-e^{-y}]_0^{\infty} = 1$$

$$\frac{k}{2} (0 - (-1)) = 1$$

$$\frac{k}{2} = 1$$

$$k = 2$$

i) Check whether x and y are independent or not.

As Joint pdf is $f(x, y) = e^{-(x+y)} = f(x) \cdot f(y)$

And marginal PDF are

$$f(x) = \int_0^{\infty} e^{-(x+y)} dy$$

$$f(y) = \int_0^{\infty} e^{-(x+y)} dx$$

Which gives

$$f(x) = -e^{-x}$$

$$f(y) = -e^{-y}$$

Thus $f(x, y) = f(x) \cdot f(y)$

Hence x and y are independent variables.

ii) Find $P(0 < x < 2, 0 < y < 5)$

$$P(0 < x < 2, 0 < y < 5) = \int_0^2 \int_0^5 e^{-(x+y)} dy dx$$

$$= (e^{-2} - 1) \cdot (e^{-5} - 1) = 0.859$$

iii) $P(0 < x < 2 / 0 < y < 5)$

$$P(0 < x < 2 / 0 < y < 5) = \frac{P(0 < x < 2, 0 < y < 5)}{P(0 < y < 5)} = \frac{0.859}{0.993} = 0.865$$

Q.4.a)	Marks of students in a University follow normal distribution with mean 45 and standard deviation 5. Then out of 750 students in the university, how many students will have marks i) Greater than 45 ii) Between 35 and 50 iii) Less than 45 iv) More than 50 Answer: Each answer carries 1 mark $Z = (X-45)/5$ i) $750 * P(X > 45) = 750 * P(Z > 0) = 750 * 0.5 = 375$ ii) $750 * P(35 < X < 50) = 750 * P(-2 < Z < 1) = 750 * [F(1) - F(-2)] = 750 * (0.8413 - 0.02275) = 613.9$ iii) $750 * P(X < 45) = 750 * P(Z < 0) = 750 * 0.5 = 375$ iv) $750 * P(X > 50) = 750 * P(Z > 1) = 750 * 0.1587 = 119$ (approx.)	4 M
Q.4.b)	Consider samples of size 20 taken from a population. Sampling yields the results with mean 250 and standard deviation 25. Then estimate the population by considering i) 90% Confidence level ii) 99% Confidence level iii) 95% Confidence level iv) Comment on the effect of confidence level on estimation iv, Comment on the effect of confidence level on estimation. Note: question has missed a word "mean" after the above highlighted word. Also missed to mention the population as "normal". Therefore it may be answered any one of the following two ways of using t distribution or z distribution. Both may be considered. Answer: Each question carries 1 mark Confidence interval $(\text{mean} \pm t^*s/\sqrt{n})$ or $(\text{mean} \pm z^*s/\sqrt{n})$ i) $(\text{mean} \pm t^*s/\sqrt{n}) = (250 \pm 1.73 * 25/\sqrt{20})$ $= (240.329, 259.671)$ (or) $(\text{mean} \pm z^*s/\sqrt{n}) = (250 \pm 1.645 * 25/\sqrt{20})$ $= (240.8042, 259.1958)$ ii) $= (250 \pm 2.86 * 25/\sqrt{20}) = (234.0121, 265.9879)$ (Or) $= (250 \pm 2.58 * 25/\sqrt{20}) = (235.5774, 264.4226)$ iii) $= (250 \pm 2.09 * 25/\sqrt{20}) = (238.3165, 261.6835)$ (Or) $= (250 \pm 1.96 * 25/\sqrt{20}) = (239.0433, 260.9567)$ iv) If the confidence level increases then the length of the interval increases.	4 M